PROMETHIUM

QC Score Residue Interaction Analysis

User Guide and Case Study: CDK2 Inhibitor Binding Analysis

QCware Corporation | Promethium Platform Documentation
2025

Executive Summary

Promethium's QC Score provides a GPU-accelerated, DFT-quality residue-resolved interaction energy decomposition for protein-ligand complexes. Unlike conventional docking scores, which collapse binding affinity to a single scalar, QC Score produces a per-residue interaction fingerprint: a vector of DFT-computed interaction energies describing precisely how each ligand engages every residue in the binding pocket.

This document describes a companion Python analysis pipeline — boltzmann_average.py and qcscore_residue_analysis.py — that transforms raw QC Score output into actionable structural insights. The pipeline performs Boltzmann ensemble averaging over multi-pose datasets, automated quality control, partial least squares (PLS) regression, residue importance ranking, ligand clustering, and activity cliff detection. All outputs are designed to be interpretable by medicinal chemists without a computational background.

A complete case study using CDK2 inhibitors demonstrates how the pipeline identifies key binding pocket residues, separates active from less-active chemotypes, and flags candidate pairs for high-resolution FSAPT (InteractionMap) follow-up — closing the loop from broad interaction profiling to atomistic mechanism.

1. Background and Motivation

1.1 From Docking Score to Interaction Fingerprint

Conventional structure-based drug design relies on a single docking score to rank compounds. While computationally efficient, this approach discards the mechanistic information embedded in the protein-ligand interface: which residues are driving binding, how different chemotypes engage the pocket, and where the activity-limiting contacts are. These questions are precisely what medicinal chemists need to guide lead optimisation.

QC Score addresses this by decomposing the total interaction energy into per-residue contributions computed at DFT quality using Promethium's GPU-accelerated fragmentation framework. For a dataset of N ligands with M binding pocket residues, QC Score produces an N x M interaction energy matrix — a rich, physically grounded descriptor space that no 2D fingerprint or force-field docking method can replicate.

1.2 The Analysis Challenge

A raw interaction energy matrix is not immediately interpretable. With 30-100 residues per pocket and potentially thousands of ligands, the challenge is to extract the following without visual inspection or manual curation:

- Which residues best discriminate active from less-active ligands?
- Do ligands fall into distinct binding mode families, or is the dataset uniform?

- Which ligand pairs show unexpectedly large activity differences despite similar interaction profiles — indicating a key contact not yet explained by the residue decomposition?

The analysis pipeline described in this document answers all three questions automatically and reproducibly.

KEY INSIGHT	QC Score interaction energies are the features; the analysis pipeline is the method for extracting residue-level SAR hypotheses from them.

2. Analysis Workflow Overview

The complete pipeline consists of two scripts that run in sequence. The first handles multi-pose ensemble averaging; the second runs all analytical steps on the resulting clean dataset.

2.1 Script 1: boltzmann_average.py

When multiple docking poses have been evaluated for the same ligand, this script collapses them into a single representative row using Boltzmann weighting over score+strain:

```
python boltzmann_average.py --config boltzmann_config.json
```

score+strain is the total energy cost of the pose — the sum of the protein-ligand interaction energy and the ligand strain penalty. This is the physically correct quantity for Boltzmann weighting: poses with both strong interactions and low strain receive high weight. At 300 K, RT ≈ 0.6 kcal/mol, meaning a pose 3 kcal/mol above the best receives weight less than 1/150. In practice this is equivalent to best-pose selection for most datasets, but the framework correctly handles near-degenerate poses by genuine averaging.

The script produces a .log file documenting every pose's weight for full auditability, and three provenance columns in the output CSV (_n_poses_valid, _n_poses_blended, _best_pose) that persist through the analysis pipeline.

2.2 Script 2: qcscore_residue_analysis.py

This script accepts the Boltzmann-averaged CSV (or a raw single-pose CSV) and runs the complete analysis pipeline:

```
python qcscore_residue_analysis.py --config analysis_config.json
```

The pipeline stages run in the following order:

1. Data audit: characterises the dataset, reports failed calculations and score statistics.
2. Clash detection: excludes ligands with positive QC Score (net repulsive interaction — a pose artifact).
3. Strain filtering: excludes high-strain-energy outliers above a configurable percentile.
4. Residue exclusion: removes capped terminal residues (computational boundary artifacts) and any user-specified backbone or peripheral residues.
5. Per-residue outlier detection: identifies ligands with anomalous interaction energies at individual residues, indicative of localised clashes that survive the global score filter.
6. Variance filtering: removes residues with near-zero variance across the ligand set (invariant contacts that carry no discriminating information).
7. PLS regression: fits a partial least squares model with leave-one-out cross-validation, producing residue importance rankings and latent variable scores.
8. Clustering: k-means with automatic k selection by silhouette score, clusters auto-labelled by mean activity.
9. Activity cliff detection: identifies ligand pairs with similar interaction fingerprints but large activity differences.

3. Automated Quality Control

A significant fraction of the analytical value in this pipeline comes from its quality control layer. Clashing and artifact-containing poses do not merely add noise — they disproportionately corrupt the variance profile of individual residues, causing those residues to dominate the PLS and clustering analyses and masking genuine pharmacophoric signals. The QC layer is therefore a prerequisite for interpretable results, not an optional pre-processing step.

3.1 Clash Detection

A positive QC Score indicates net repulsive protein-ligand interaction: the ligand is pushing against the protein rather than binding. This is always a pose artifact — arising from incomplete declashing during docking, erroneous protonation states, or atom type misassignment — and is never physically meaningful as a binding affinity surrogate. All such ligands are excluded unconditionally and named in the audit report with their score values.

NOTE	The clash_score_threshold parameter (default: 0 kcal/mol) can be adjusted if a slightly positive score is expected for a specific system, but the default of zero is appropriate for the vast majority of use cases.

3.2 Per-Residue Outlier Detection

A ligand may have a plausible total QC Score while still containing a localised clash at one residue — for example, a pose that engages the core of the binding pocket correctly but clips a peripheral sidechain. These localised artifacts are detected by flagging residue interaction values beyond ±N×IQR from the dataset median for each residue. The default factor of 10× is deliberately conservative, catching only genuine artifacts rather than real pharmacophoric variation. Flagged ligands and their offending residues are reported in the audit file.

3.3 Capped Residue Exclusion

QC Score decomposes interaction energies using an alanine-scanning fragmentation scheme. The terminal caps of each peptide fragment (identifiable by parentheses in the residue name, e.g. (V7A)) are computational boundary artifacts: their interaction energies partially reflect fragmentation edge effects rather than genuine ligand contacts. These residues are excluded by default (exclude_capped_residues: true) before any modelling.

4. Analytical Methods

4.1 Partial Least Squares Regression

Partial least squares (PLS) is the primary analytical method for identifying activity-relevant residues. Unlike principal component analysis (PCA), which finds axes of maximum variance regardless of their relationship to activity, PLS finds linear combinations of residue interactions that maximally co-vary with the activity axis (QC Score or score+strain). The resulting latent variable loadings directly rank residues by their contribution to the activity-discriminating axis.

The optimal number of PLS components is selected by leave-one-out (LOO) cross-validation for datasets under 30 ligands, and 10-fold cross-validation for larger datasets. The cross-validated R^2 (Q^2) is reported as the primary model quality metric.

IMPORTANT	A near-zero CV R^2 does not mean the analysis is uninformative. With fewer than ~30 ligands, LOO prediction is unreliable regardless of signal quality. The PLS loadings and Spearman correlations remain valid descriptive statistics for residue hypothesis generation. Predictive modelling becomes reliable above approximately 50-100 ligands.

4.2 Spearman Rank Correlation

In parallel with PLS, the Spearman rank correlation between each residue's interaction energies and the activity axis is computed across the ligand set. Spearman correlation is more robust than Pearson correlation for small, potentially non-linear datasets and is the primary statistic for residue prioritisation at small sample sizes. A negative Spearman ρ means that stronger interaction with that residue (more negative interaction energy) co-varies with better binding — the expected relationship for pharmacophoric contacts. Positive ρ residues, where stronger interaction correlates with weaker binding, may indicate gating residues, non-productive binding modes, or competitive contacts worth investigating.

4.3 K-means Clustering

K-means clustering partitions ligands by similarity in residue interaction space, identifying groups that engage the pocket in qualitatively similar ways. The number of clusters k is selected automatically by maximising the silhouette score over the configured search range. Clusters are automatically labelled by mean activity rank (cluster 0 = most active) without requiring visual inspection of box plots, eliminating a key manual step from the original analysis workflow.

The resulting clusters should be interpreted as interaction fingerprint families, not chemical scaffold families — two structurally diverse ligands with similar binding modes will cluster together, and two analogues with different binding orientations will separate. This distinction is often more informative for lead optimisation than scaffold-based clustering.

4.4 Activity Cliff Detection

Activity cliffs are identified as ligand pairs that are similar in residue interaction space but show large differences in activity. The cliff score is defined as:

```
cliff_score = normalised_activity_difference / (normalised_interaction_distance
+ ε)
```

High cliff score pairs are the most valuable candidates for InteractionMap (FSAPT) follow-up: they represent cases where the gross residue interaction profile does not explain the activity difference, pointing to a specific contact — or its decomposition into electrostatic, dispersion, and induction components — that requires higher-resolution analysis.

Note that in the absence of SMILES data, interaction-space distance is used as a proxy for structural similarity. Adding a cross-reference file with SMILES enables Tanimoto-filtered cliff detection, which is more rigorous.

5. Configuration Reference

5.1 boltzmann_config.json

Parameter	Default	Description
`input_file`	qcscore_output.csv	Path to raw QC Score CSV from Promethium
`output_file`	qcscore_boltzmann.csv	Path for Boltzmann-averaged output CSV

`boltzmann_temperature`	300	Temperature in K for Boltzmann weighting. At 300 K, RT ≈ 0.6 kcal/mol (best-pose selection). Increase for softer averaging.
`clash_score_threshold`	0	Poses with score+strain above this value are excluded as clashing before weighting.
`pose_suffix_pattern`	_[0-9]+$	Regex pattern identifying pose suffixes. Default matches _1, _2, _3 etc.

5.2 analysis_config.json

Parameter	Default	Description
`data_file`	qcscore_output.csv	Input CSV. Use boltzmann output for multi-pose datasets.
`activity_column`	score	Activity axis: score or score_plus_strain. Run both if audit reports Spearman ρ < 0.95 between them.
`activity_higher_is_better`	true	Sign convention. True for score/score_plus_strain (script flips sign internally so more negative = higher y).
`clash_score_threshold`	0	Ligands with activity column above this threshold are excluded as clashing.
`residue_iqr_outlier_factor`	10.0	Per-residue outlier sensitivity. Lower = stricter. Set null to disable.
`strain_filter_percentile`	95	Ligands above this strain percentile are excluded. Set null to disable.
`exclude_capped_residues`	true	Drop residues in parentheses (computational boundary artifacts). Recommended: true.
`exclude_residues`	[]	Explicit list of residue column names to exclude (e.g. backbone/bridging residues).
`variance_filter_percentile`	10	Drop residues below this variance percentile. Increase to 20-25 for large diverse sets.
`n_pls_components`	10	Maximum PLS components to evaluate. Optimal value selected by CV.
`n_clusters`	null	null = auto-select by silhouette. Integer = fix k.
`k_range`	[2, 20]	Search range for k. Recommended: [2,10] for <100 ligands; [2,20] for 100-3000.
`cliff_sample_size`	null	null = use all ligands (up to ~5000). Set integer to cap for large datasets.
`output_dir`	qcscore_analysis_output	Directory for all output files.

6. Output Files

All outputs are written to the configured output_dir. The most important for a first-pass analysis are residue_ranking.csv, pls_residue_loadings.png, and activity_cliffs.csv.

Parameter	Default	Description
`audit_report.txt`	—	Data quality report: failed calculations, clash exclusions, outlier exclusions, strain filter, variance profile, and score vs score+strain correlation.
`summary_report.txt`	—	Plain-language findings: ligand accounting, PLS model quality, clustering result, top 10 residues, and recommended next steps.
`residue_ranking.csv`	—	Full ranked residue table: PLS LV1 loading, LV2 loading, Spearman ρ, p-value. Primary output for medicinal chemistry follow-up.
`residue_variance.png`	—	Ranked residue variance profile. Useful for assessing feature space quality and variance filter threshold.
`pls_cv_scores.png`	—	Cross-validated R^2 vs number of PLS components. Assesses model predictive quality and dimensionality.
`pls_residue_loadings.png`	—	Side-by-side bar charts: PLS LV1 loadings (left) and Spearman ρ (right) for top N residues. Primary visualisation.
`pls_scores_scatter.png`	—	Ligand positions in PLS LV1/LV2 space, coloured by activity. Reveals binding mode structure.
`cluster_silhouette.png`	—	Silhouette score vs k. Shows quality of automatic k selection.
`cluster_pca_scatter.png`	—	Cluster membership overlaid on PCA projection of residue interaction space.
`cluster_activity_whisker.png`	—	Activity distribution per cluster. Confirms clusters stratify by binding quality.
`activity_cliffs.csv`	—	Top 50 activity cliff pairs ranked by cliff score. Primary list for FSAPT follow-up prioritisation.

7. Case Study: CDK2 Inhibitor Analysis

Cyclin-dependent kinase 2 (CDK2) is a well-characterised drug target with a large body of structural and biochemical data, making it an ideal validation system for the analysis pipeline. This case study demonstrates the full workflow from raw multi-pose QC Score output to actionable residue hypotheses, using a set of 16 CDK2 inhibitors with known crystal structures.

7.1 Dataset Preparation

The raw QC Score dataset comprised 80 rows: 16 ligands each evaluated in one original crystallographic pose and up to three docked poses, plus a small number of duplicate failed calculation entries. Boltzmann ensemble averaging was applied first to collapse the multi-pose data to one representative row per ligand.

Boltzmann Averaging Result

At 300 K, 14 of 16 ligands were effectively single-pose (dominant pose weight > 0.99). Only ligand_29, whose two valid docked poses differed by less than 0.1 kcal/mol in score+strain, showed genuine blending (effective poses = 1.99). This result is physically expected: docked poses typically differ by 5-15 kcal/mol, giving Boltzmann weight ratios of 10^4 to 10^{11}. The Boltzmann framework correctly reduces to best-pose selection in this regime.

Quality Control Results

After Boltzmann averaging, the analysis pipeline applied the following quality filters:

Filter	Excluded	Notes
Clashing (score > 0)	0	All ligands had negative QC Score after Boltzmann averaging.
Per-residue outlier (10×IQR)	1	ligand_1oiy flagged for anomalous N132 interaction energy.
High strain (>95th percentile)	1	One ligand excluded for strain energy above threshold.
Retained for modelling	14	Clean dataset for PLS and clustering.

7.2 Residue Variance Profile

The residue variance profile reveals the information content of the interaction fingerprint. A healthy profile shows a gradual decay from high-variance to low-variance residues, indicating that multiple residues contribute discriminating information. The CDK2 dataset (46 residues after cap exclusion) shows this expected pattern, with meaningful variance distributed across approximately the top 20 residues before tapering to near-zero.

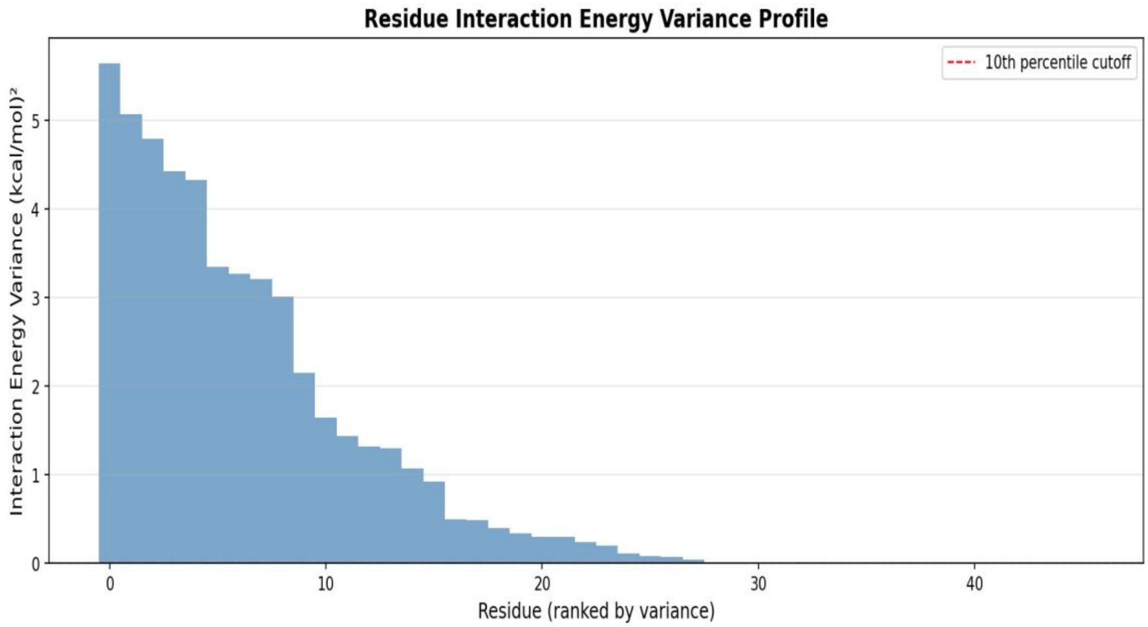

Figure 1. Residue interaction energy variance profile after capped residue exclusion. The gradual decay indicates variance is distributed across multiple informative residues, not dominated by a single artifact.

7.3 PLS Model Quality

Cross-validated R^2 (Q^2) is near zero for the 14-ligand CDK2 set, which is expected for LOO cross-validation at small n with a high-dimensional feature space (41 residues). This reflects a sample size limitation, not the absence of signal. The PLS loadings and Spearman correlations below are statistically valid descriptive statistics and are the correct outputs to use for residue hypothesis generation at this scale.

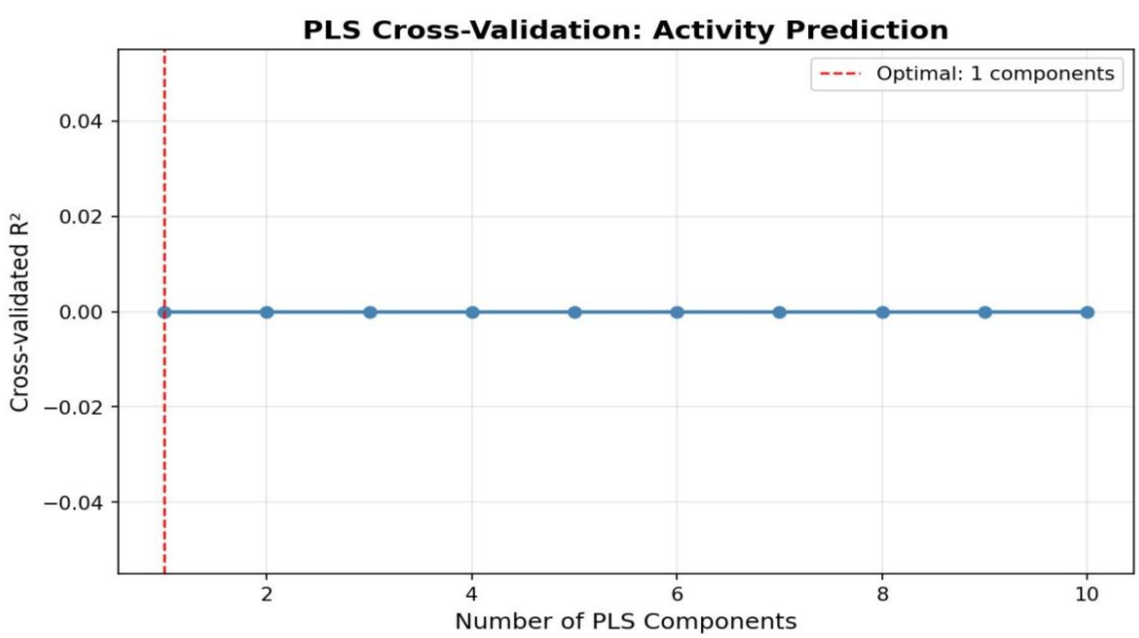

Figure 2. LOO cross-validated R² vs number of PLS components. Near-zero Q² is expected at n=14. The PLS loadings remain informative as descriptive statistics.

7.4 Residue Importance Ranking

The residue loading and Spearman correlation plot is the primary analytical output. The left panel shows PLS LV1 loadings (the contribution of each residue to the activity-discriminating axis); the right panel shows Spearman ρ between each residue's interaction energies and QC Score across the 14 ligands. Both statistics are presented side-by-side because their agreement provides confidence in the finding.

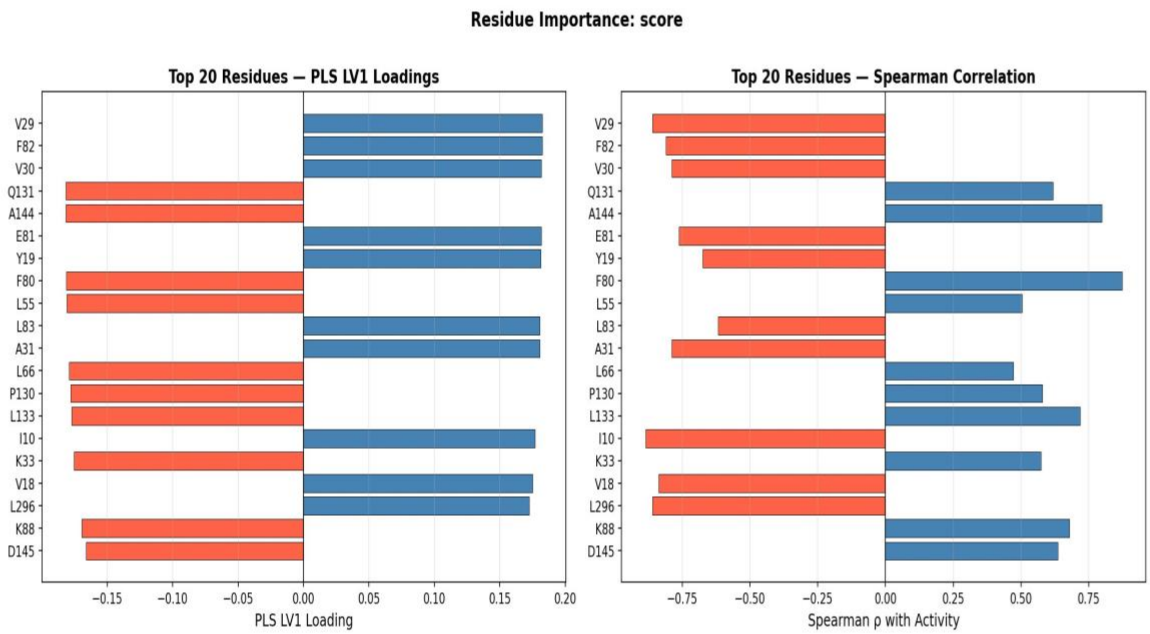

Figure 3. Top 20 residues by PLS LV1 loading (left) and Spearman rank correlation with QC Score (right). Blue bars indicate residues where stronger interaction co-varies with better binding (negative ρ); orange bars indicate the opposite. Strong sign agreement between the two panels confirms robustness of the ranking.

Key findings from the residue ranking:

- V29, F82, V30, E81, Y19, I10 (negative ρ = 0.86 to 0.76): Stronger interaction with these residues co-varies with better QC Score. These are primary pharmacophoric contacts for CDK2 inhibition and consistent with known ATP-binding site pharmacophore features.
- F80, A144, Q131, L133 (positive ρ = 0.87 to 0.72): Stronger interaction with these residues co-varies with weaker binding. Residues with positive ρ can indicate competitive contacts, non-productive binding orientations, or gating residues that constrain ligand access. F80 in particular is a known gating residue in CDK2.
- I10 at $\rho = -0.89$ ($p = 2.5\times10^{-5}$): The strongest Spearman correlation in the dataset and statistically highly significant, identifying I10 as the most consistent pharmacophoric contact across this ligand series.

7.5 Ligand Clustering

K-means clustering with automatic k selection identified k=2 as optimal (silhouette score = 0.60), reflecting a clean separation of the 14-ligand set into two interaction fingerprint families.

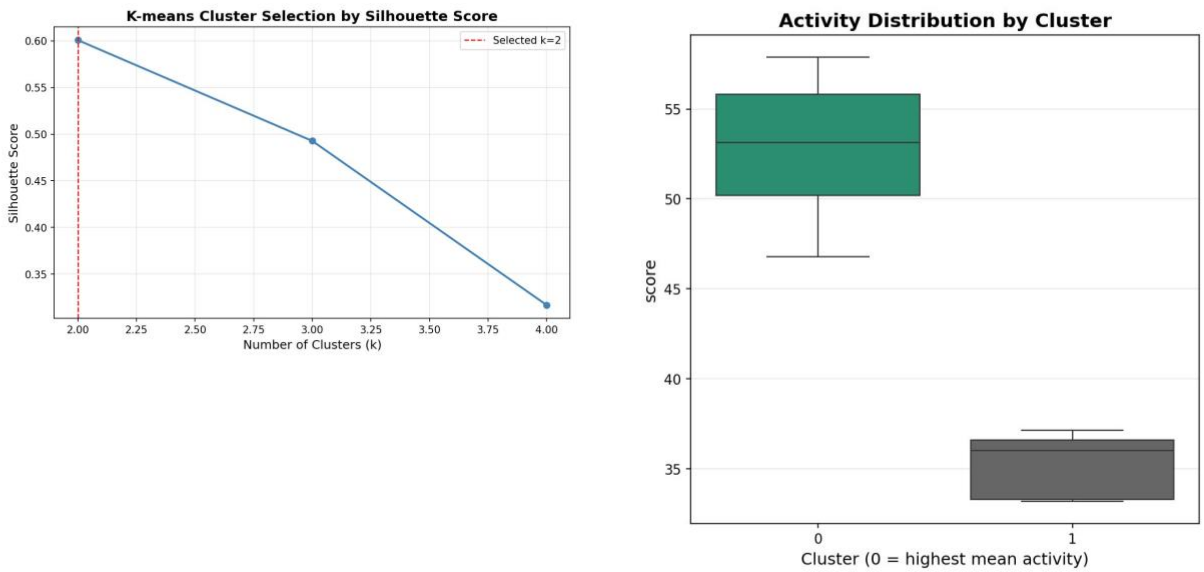

Figure 4. Left: silhouette score vs k showing a clear optimum at k=2. Right: activity distribution by cluster. Cluster 0 (mean QC Score magnitude 52.7 kcal/mol) and cluster 1 (35.3 kcal/mol) show non-overlapping IQRs, confirming meaningful biological stratification.

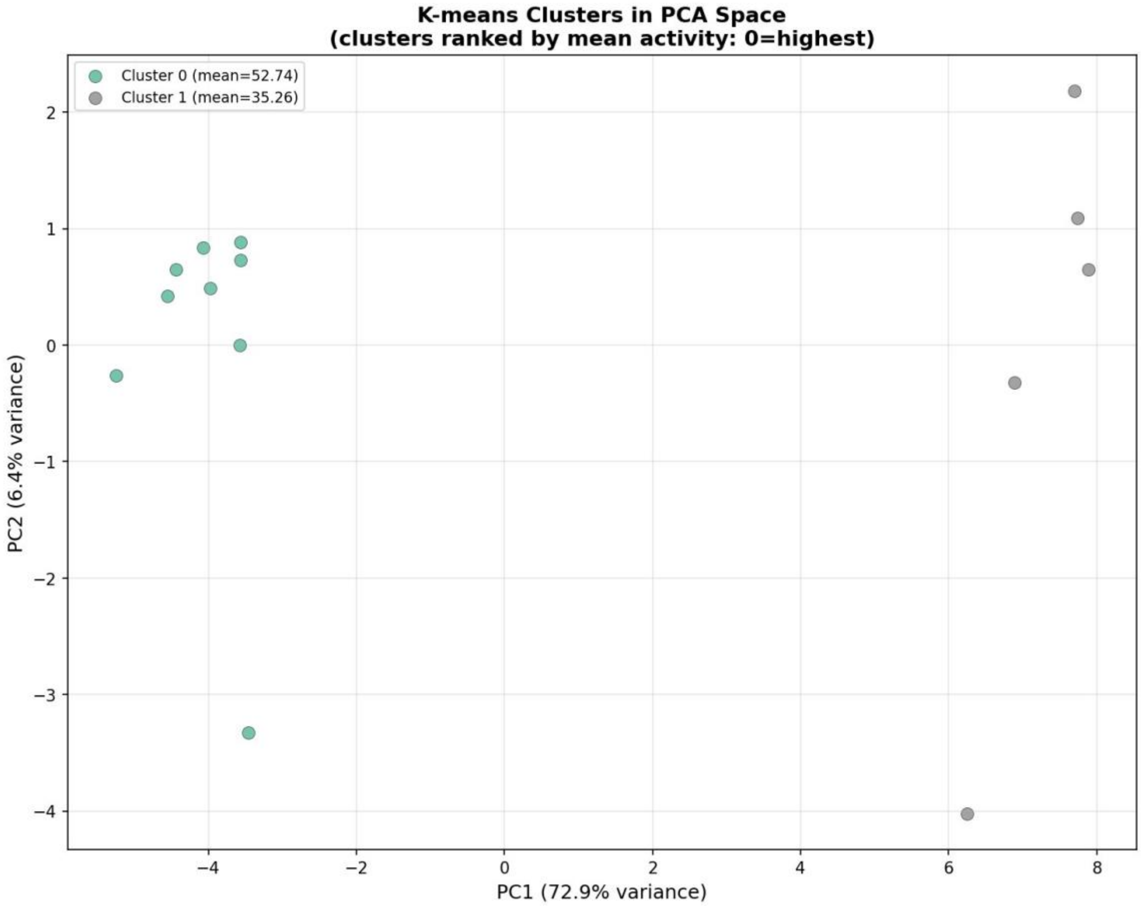

Figure 5. K-means cluster membership in PCA space. PC1 accounts for 72.9% of variance, reflecting the dominant one-dimensional structure of this congeneric ligand set. The two clusters are well-separated along PC1.

The non-overlapping activity distributions in the whisker plot confirm that the two clusters represent genuinely different binding quality groups, not arbitrary geometric partitions. This separation was identified automatically from the interaction fingerprint without any activity threshold being specified.

7.6 Activity Cliff Detection

The top activity cliff pairs identify ligands that are similar in interaction fingerprint space but show the largest differences in QC Score. These are the highest-value candidates for InteractionMap (FSAPT) follow-up, which decomposes residue interactions into electrostatic, exchange-repulsion, induction, and dispersion components.

Top three activity cliff pairs:

- ligand_1oiu vs ligand_21: interaction distance 4.61, activity difference 10.4 kcal/mol, cliff score 1.27. These two ligands engage the pocket in similar ways but differ significantly in total binding energy — a prime candidate for FSAPT to identify the contact responsible for the gap.

- ligand_21 vs ligand_26: interaction distance 3.86, activity difference 8.3 kcal/mol, cliff score 1.21. Near-degenerate interaction profiles with a non-trivial activity gap.
- ligand_20 vs ligand_29: interaction distance 11.66, activity difference 24.2 kcal/mol, cliff score 1.17. Larger structural difference but enormous activity gap — likely binding mode distinction worth structural investigation.

RECOMMENDED ACTION	Run InteractionMap (FSAPT) on the top 3-5 cliff pairs from activity_cliffs.csv. Focus on residues in the top 5 of the Spearman ranking. This combination of broad QC Score profiling followed by targeted FSAPT is the intended analysis workflow and typically reveals the atomistic mechanism of the activity difference within 1-2 calculation cycles.

8. Interpreting Analysis Outputs

8.1 What CV R^2 Near Zero Means

The cross-validated R^2 is a measure of whether the PLS model can predict activity for held-out ligands. For small datasets (<30 ligands), LOO prediction is inherently unreliable: with only one hold-out observation per fold, prediction accuracy is dominated by sampling variance rather than model quality. A Q^2 near zero in this regime should be interpreted as 'dataset too small for predictive modelling', not as 'residue interactions do not correlate with activity'.

The PLS loadings and Spearman correlations, by contrast, are computed over the full dataset without hold-out and remain valid descriptive statistics at any sample size. They answer the question: across this ligand set, which residue interaction energies co-vary with activity? This is precisely the question needed for residue hypothesis generation, and it does not require predictive power.

8.2 What LV2 Collapse Means

When only one PLS latent variable is found to be meaningful (LV2 variance near zero), the residue interaction space is essentially one-dimensional for the ligand set being analysed. All ligands vary along a single axis of interaction pattern differences. This is common for congeneric series or small sets of structurally similar ligands engaging the pocket in the same qualitative mode. LV2 collapse simplifies interpretation: a single ranked list of residues fully describes the SAR structure of the dataset.

When LV2 is non-collapsed and CV R^2 is low simultaneously, this suggests binding mode heterogeneity — the ligand set contains sub-populations engaging the pocket in fundamentally different ways, which suppresses the global model. In this case, running the analysis separately per cluster is recommended.

8.3 Positive Spearman ρ Residues

Residues with positive Spearman ρ (stronger interaction correlating with weaker binding) are often the most pharmacologically interesting finding. Possible interpretations include:

- Gating or selectivity residues: contacts that are structurally required for access to the binding pocket but whose over-engagement is associated with sub-optimal poses.
- Non-productive binding mode markers: ligands that engage these residues may be binding in an orientation that simultaneously compromises a key pharmacophoric contact.
- Allosteric or induced-fit effects: strong interaction with a peripheral residue may indicate conformational strain being transmitted to the active site.

These residues should not be dismissed as noise. FSAPT decomposition of the interaction at a positive-ρ residue often reveals an unfavourable electrostatic or exchange-repulsion component that explains the anti-correlation.

8.4 Silhouette Curve Interpretation

A healthy silhouette curve for a real dataset shows a clear peak at the optimal k, declining on both sides. A monotonically decreasing curve (k=2 always wins) typically indicates the ligands form one homogeneous population or that only two distinct binding quality groups are present — which is often biologically meaningful for a congeneric series. A monotonically increasing curve hitting the k_range ceiling indicates the dataset geometry is dominated by outliers or that the feature space is too noisy for meaningful clustering; the audit and outlier filters should be reviewed.

9. Scaling to Large Datasets

The pipeline is designed to scale from small proof-of-concept datasets (10-30 ligands) to production datasets of thousands of ligands without code changes. The following parameters should be adjusted for large datasets:

Parameter	Default	Description
`k_range`	[2, 20] or wider	For 500+ ligands, extend upper bound to 25-30. Silhouette evaluation is fast even at k=30 for large n.
`variance_filter_percentile`	15–25	For diverse large datasets, more residues may show meaningful variance. Raising this threshold focuses analysis on the highest-signal contacts.
`cliff_sample_size`	null up to ~5000	Pairwise distance computation scales as n^2. For n > 5000, set cliff_sample_size to a stratified subsample (e.g. 2000). For n < 5000, leave as null.
`n_pls_components`	10–15	With 500+ ligands, more PLS components may be justified. The CV curve will show a clear optimum.

For datasets above 50 ligands, the CV R^2 becomes a meaningful quality metric and the Spearman correlations achieve statistical power to support confident residue prioritisation. The combination of strong CV R^2 and consistent Spearman ρ at this scale is sufficient for QSAR-quality residue importance ranking.

10. Quick Start

To run the pipeline on a new QC Score dataset:

10. Export the QC Score CSV from Promethium.

11. If multiple poses per ligand were evaluated, run Boltzmann averaging first:

```
python boltzmann_average.py --config boltzmann_config.json
```

12. Edit analysis_config.json to point to your input file and set output_dir.
13. Run the analysis:

```
python qcscore_residue_analysis.py --config analysis_config.json
```

14. Review audit_report.txt first. Check that the number of excluded ligands is reasonable and that no unexpected residues were flagged.
15. Review summary_report.txt for the top 10 residues and model quality metrics.
16. Open pls_residue_loadings.png for the primary visualisation.
17. Review activity_cliffs.csv for FSAPT follow-up candidates.

TIP	Run the analysis twice — once with activity_column: score and once with activity_column: score_plus_strain. If the audit reports Spearman ρ between the two below 0.95, the residue rankings may differ meaningfully. Compare the top-10 lists from both runs to identify contacts that are robust regardless of which activity axis is used.

Appendix: Installation and Dependencies

Both scripts require Python 3.8+ with the following packages:

```
pip install pandas numpy scipy scikit-learn matplotlib seaborn
```

No additional Promethium credentials or API access are required to run the analysis scripts. They operate entirely on the exported CSV file.

The scripts have been tested on macOS, Linux, and Windows with Python 3.9-3.12. The boltzmann_average.py output CSV is directly compatible with qcscore_residue_analysis.py without any format conversion.

For questions about the analysis pipeline or to report issues, contact your QCware customer success representative.